

CSCI 6840

Project 3

Convolutional Neural Networks (CNNs) for Image Classification on the CIFAR-10 Dataset

Objective:

- Build a Convolutional Neural Network (CNN) to classify images in the CIFAR-10 dataset.
- Apply data augmentation to improve model generalization.
- Tune hyperparameters to minimize overfitting and underfitting, and to achieve the highest possible classification accuracy.
- Train and evaluate the model on the CIFAR-10 dataset.

Skills Covered:

- Python and PyTorch programming
- CNN architectures
- Hyperparameter tuning (including but not limited to CNN architecture, learning rate, batch size, number of epochs)
- Techniques for reducing overfitting (e.g., dropout, data augmentation)
- Model evaluation

Required Tasks:

1. Dataset loading and visualization
2. Building a CNN model
3. Implementing image data augmentation
4. Hyperparameter tuning (including but not limited to CNN architecture, learning rate, batch size, number of epochs) to prevent overfitting/underfitting, and to achieve the highest possible classification accuracy.
5. Model training and evaluation
6. Preventing overfitting/underfitting
7. Reporting results and analyzing model performance

Deliverables:

- Python code in .py file.
- A report discussing model design, hyperparameter tuning, and results.
- Presentation slides. In your presentation, you are required to explain your code.

Notes:

- Your implementation should be in PyTorch.
- Do NOT use other deep learning models. Just use CNN models (convolution layers, pooling, activation functions, etc.).

Grading Metric:

1. Dataset loading and visualization (10%)
2. Building a CNN model (30%)
3. Implementing image data augmentation (10%)
4. Hyperparameter tuning (including but not limited to CNN architecture, learning rate, batch size, number of epochs) to prevent overfitting/underfitting, and to achieve the highest possible classification accuracy. (25%)
5. Model training and evaluation (15%)
6. Reporting results and analyzing model performance (10%)